

Chapter C5: Chemical analysis

Overview

This chapter looks at how chemicals are analysed. Chemical analysis is important in chemistry for the quality control of manufactured products and also to identify or quantify components in testing of new products, mineral extraction, forensics and environmental monitoring. Chemists need to both identify which substances are present (qualitative analysis) and the quantity of each substance (quantitative analysis). Measuring purity and separating mixtures is important in manufacturing to ensure quality and to separate useful products from by-products and waste. Being able to analyse quantities of chemicals enables chemists to plan for the amounts of reactants they need to use to make a product, or predict quantities of products from known amounts of reactants.

The chapter begins in Topic C5.1 by considering why it is necessary to purify chemicals and how the components of mixtures are separated. Methods of testing for purity and separating mixtures are studied, including chromatography and a range of practical separation techniques.

Topic C5.2 is concerned with qualitative analysis. This topic uses standard laboratory techniques, such as flame tests, precipitation reactions and anion tests to identify the ions in unknown substances. Gas chromatography is introduced as an example of an instrumental technique, and used as a context for comparison between standard laboratory and instrumental techniques.

Topic C5.3 introduces quantitative work. The mole is used as a measure of amounts of substance and learners process data from formulae and equations to work out quantities of reactants and products. This topic ends by considering how molar amounts of gas relate to their volumes.

Topic C5.4 develops quantitative work further to show how the concentrations of solutions are determined. This has applications for the testing and quality control of manufactured chemical products and also allows the analysis of unknown chemicals for a range of purposes (for example in forensics, in drug production, mineral exploration and environmental monitoring). Learners make a standard solution and analyse the concentration of unknown solutions using titrations.

Learning about Chemical analysis before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- use knowledge of solids, liquids and gases to decide how mixtures might be separated, including through filtering, sieving and evaporating
- understand the concept of a pure substance and how to identify a pure substance
- know about simple techniques for separating mixtures: filtration, evaporation, distillation and chromatography
- know about the pH scale for measuring acidity/alkalinity; and indicators.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about Chemical Analysis at GCSE (9–1)

C5.1 How are chemicals separated and tested for purity?	
Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Many useful products contain mixtures. It is important that consumer products such as drugs or personal care products do not include impurities. Mixtures in many consumer products contain pure substances mixed together in definite proportions called formulations. Pure substances contain a single element or compound. Chemists test substances made in the laboratory and in manufacturing processes to check that they are pure. One way of assessing the purity of a substance is by testing its melting point; pure substances have sharp melting points and can be identified by matching melting point data to reference values. Chromatography is used to see if a substance is pure or to identify the substances in a mixture. Components of a mixture are identified by the relative distance travelled compared to the distance travelled by the solvent. Rf values can be calculated and used to identify unknown components by comparison to reference samples. Some substances are insoluble in water, so other solvents are used. Chromatography can be used on colourless substances but locating agents are needed to show the spots. Preparation of chemicals often produces impure products or a mixture of products. Separation processes in both the laboratory and in industry enable useful products to be separated from by-products and waste products. The components of mixtures are separated using processes that exploit the different properties of the components, for example state, boiling points or solubility in different solvents. Separation processes are rarely completely successful and mixtures often need to go through several stages or through repeated	1. explain that many useful materials are formulations of mixtures
	2. explain what is meant by the purity of a substance, distinguishing between the scientific and everyday use of the term 'pure'
	3. use melting point data to distinguish pure from impure substances
	4. recall that chromatography involves a stationary and a mobile phase and that separation depends on the distribution between the phases
	5. interpret chromatograms, including calculating Rf values M3c
	6. suggest chromatographic methods for distinguishing pure from impure substances PAG4
	Including the use of: a) paper chromatography b) aqueous and non-aqueous solvents c) locating agents
	7. describe, explain and exemplify the processes of filtration, crystallisation, simple distillation, and fractional distillation PAG3, PAG7
	8. suggest suitable purification techniques given information about the substances involved PAG3, PAG7

Linked learning opportunities

Specification links:

- particle model and changes of state (C1.1)
- Fractional distillation of crude oil on an industrial scale (C3.4)

Ideas about Science

- use the particle model to explain the idea of a pure substance